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*
*          STAAD.Pro
*          Version  2007    Build 04
*          Proprietary Program of
*          Research Engineers, Intl.
*          Date=    MAY 14, 2013
*          Time=    15:29:39
*
*          USER ID: Parsons Brinckerhoff
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1. STAAD SPACE
INPUT FILE: 130503-ST28-SCQ-0557 + 0448.STD
2. START JOB INFORMATION
3. ENGINEER DATE 06-JAN-12
4. END JOB INFORMATION
5. INPUT WIDTH 79
6. UNIT METER KN
7. JOINT COORDINATES
8. 2 1.0004 -3 -9.98974E-005; 5 1.0004 -3 -7.2001; 7 6.0004 -3 -9.98974E-005
9. 8 13.0004 -3 -9.98974E-005; 9 19.0004 -3 -9.98974E-005; 13 6.0004 -3 -7.2001
10. 18 13.0004 -3 -7.2001; 23 19.0004 -3 -7.2001; 25 19.0004 2.99982 -5.7001
11. 27 13.0004 2.99982 -5.7001; 29 6.0004 2.99982 -5.7001
12. 31 1.0004 2.99982 -5.7001; 35 1.0004 1.49971 -9.98974E-005
13. 36 1.0004 1.49971 -7.2001; 39 6.0004 1.49971 -9.98974E-005
14. 40 6.0004 1.49971 -7.2001; 43 13.0004 1.49971 -9.98974E-005
15. 44 13.0004 1.49971 -7.2001; 47 19.0004 1.49971 -9.98974E-005
16. 48 19.0004 1.49971 -7.2001; 49 19.0004 2.99982 -1.5001
17. 51 13.0004 2.99982 -1.5001; 53 6.0004 2.99982 -1.5001
18. 55 1.0004 2.99982 -1.5001; 1000 1.0004 0.99971 -7.2001
19. 1001 1.0004 0.99971 -9.98974E-005; 1002 1.0004 2.99982 -7.2001
20. 1003 1.0004 2.99982 -9.98974E-005; 1004 6.0004 0.99971 -7.2001
21. 1005 6.0004 0.99971 -9.98974E-005; 1006 6.0004 2.99982 -7.2001
22. 1007 6.0004 2.99982 -9.98974E-005; 1008 13.0004 0.99971 -7.2001
23. 1009 13.0004 0.99971 -9.98974E-005; 1010 13.0004 2.99982 -7.2001
24. 1011 13.0004 2.99982 -9.98974E-005; 1012 19.0004 0.99971 -7.2001
25. 1013 19.0004 0.99971 -9.98974E-005; 1014 19.0004 2.99982 -7.2001
26. 1015 19.0004 2.99982 -9.98974E-005; 1018 14.5004 2.99982 -9.98974E-005
27. 1019 17.5004 2.99982 -9.98974E-005
28. MEMBER INCIDENCES
29. 1 2 1001; 2 5 1000; 3 1001 35; 4 1000 36; 5 7 1005; 6 13 1004; 7 1005 39
30. 8 1004 40; 9 8 1009; 10 18 1008; 11 1009 43; 13 9 1013; 14 23 1012; 15 1013 47
31. 17 1002 1006; 18 1006 1010; 19 1010 1014; 20 1003 1007; 21 1007 1011
32. 22 1011 1018; 23 1001 1005; 24 1005 1009; 25 1009 1013; 26 1000 1004
33. 27 1004 1008; 28 1008 1012; 29 1014 25; 30 1012 1013; 31 1010 27; 32 1008 1009
34. 33 1006 29; 34 1004 1005; 35 1002 31; 36 1000 1001; 37 2 1005; 38 1001 7
35. 39 1004 5; 40 1000 13; 41 18 1012; 45 25 49; 47 27 51; 49 29 53; 51 31 55
36. 55 35 1003; 56 36 1002; 59 39 1007; 60 40 1006; 63 43 1011; 64 44 1010
37. 67 47 1015; 68 48 1014; 69 49 1015; 71 51 1011; 73 53 1007; 75 55 1003
38. 78 36 31; 79 40 29; 80 44 27; 81 48 25; 83 35 55; 85 39 53; 87 43 51; 89 47 49
39. 97 1018 1019; 98 1019 1015; 99 1018 43; 100 1019 47; 103 1008 44; 104 1012 48
40. 105 1008 23
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41. DEFINE MATERIAL START
42. ISOTROPIC STEEL
43. E 2.05E+008
44. POISSON 0.3
45. DENSITY 76.8195
46. ALPHA 1.2E-005
47. DAMP 0.03
48. END DEFINE MATERIAL
49. MEMBER PROPERTY JAPANESE
50. 1 TO 11 13 TO 15 55 56 59 60 63 64 67 68 103 104 TABLE ST H394X398X11
51. 37 TO 41 105 TABLE ST C150X75X6.5
52. 17 TO 36 45 47 49 51 69 71 73 75 97 98 TABLE ST H394X398X11
53. 78 TO 81 83 85 87 89 99 100 TABLE ST H200X200X8
54. CONSTANTS
55. BETA 90 MEMB 1 TO 11 13 TO 15 55 56 59 60 63 64 67 68 103 104
56. MATERIAL STEEL ALL
57. SUPPORTS
58. 2 5 7 8 13 18 23 PINNED
59. 9 FIXED BUT MX MY KMZ 0.001
60. MEMBER RELEASE
61. 17 TO 22 25 28 START MY MZ
62. 17 TO 21 25 28 98 END MY MZ
63. 78 TO 81 83 85 87 89 START MY MZ
64. 78 TO 81 83 85 87 89 END MY MZ
65. 100 START MY MZ
66. 100 END MY MZ
67. 99 START MY MZ
68. 99 END MY MZ
69. 23 24 26 27 START MY MZ
70. 23 24 26 27 END MY MZ
71. *30 32 34 36 START MPZ 0.3
72. *30 32 34 36 END MPZ 0.3
73. MEMBER TENSION
74. 37 TO 41 105
75. LOAD 1 LOADTYPE NONE TITLE DL
76. SELFWEIGHT Y -1
77. MEMBER LOAD
78. 30 UNI GY -29.98
79. 36 UNI GY -20.18
80. 34 UNI GY -20.18
81. 27 UNI GY -12.95
82. 29 45 69 UNI GY -32.09
83. 31 47 71 UNI GY -32.95
84. 33 35 49 51 73 75 UNI GY -63.81
85. 17 UNI GY -59.38
86. 32 UNI GY -30.07
87. LOAD 2 LOADTYPE NONE TITLE FL
88. MEMBER LOAD
89. 30 UNI GX -3.15
90. 32 UNI GX -3.51
91. 36 UNI GX -3.51
92. 34 UNI GX -3.51
93. 27 UNI GZ -1.3
94. 29 31 45 47 69 71 UNI GX -3.32
95. 33 35 49 51 73 75 UNI GX -6.38
96. 17 UNI GZ -17.82
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97. LOAD 3 LOADTYPE NONE TITLE WLX
98. MEMBER LOAD
99. 1 TO 11 13 TO 15 29 TO 31 33 35 45 47 49 51 55 56 59 60 63 64 67 TO 69 71 -
100. 73 75 103 104 UNI GX -0.34
101. 36 UNI GX -0.34
102. 34 UNI GX -0.34
103. 32 UNI GX -0.34
104. 78 TO 81 83 85 87 89 99 100 UNI GX -0.17
105. LOAD 4 LOADTYPE NONE TITLE WLZ
106. MEMBER LOAD
107. 1 TO 11 13 TO 15 17 TO 22 25 28 TO 31 33 35 37 TO 41 45 47 49 51 55 56 59 -
108. 60 63 64 67 TO 69 71 73 75 97 98 103 TO 105 UNI GZ -0.34
109. 23 UNI GZ -0.34
110. 36 UNI GZ -0.34
111. 27 UNI GZ -0.34
112. 24 UNI GZ -0.34
113. 26 UNI GZ -0.34
114. 34 UNI GZ -0.34
115. 32 UNI GZ -0.34
116. 37 TO 41 105 UNI GZ -0.13
117. 78 TO 81 83 85 87 89 99 100 UNI GZ -0.17
118. LOAD 5 LOADTYPE NONE TITLE NLX
119. SELFWEIGHT X -0.015
120. MEMBER LOAD
121. 30 UNI GX -0.45
122. 32 UNI GX -0.45
123. 36 UNI GX -0.3
124. 34 UNI GX -0.3
125. 27 UNI GX -0.19
126. 29 45 69 UNI GX -0.48
127. 31 47 71 UNI GX -0.49
128. 33 35 49 51 73 75 UNI GX -0.96
129. 17 UNI GX -0.89
130. LOAD 6 LOADTYPE NONE TITLE NLZ
131. SELFWEIGHT Z -0.015
132. MEMBER LOAD
133. 30 UNI GZ -0.45
134. 32 UNI GZ -0.45
135. 36 UNI GZ -0.3
136. 34 UNI GZ -0.3
137. 27 UNI GZ -0.19
138. 29 45 69 UNI GZ -0.48
139. 31 47 71 UNI GZ -0.49
140. 17 UNI GZ -0.9
141. 17 UNI GZ -0.89
142. 33 35 49 51 73 75 UNI GZ -0.96
143. LOAD COMB 7 1.0DL+1.0FL
144. 1 1.0 2 1.0
145. LOAD COMB 8 1.0DL+0.80 (FL+ WLX)
146. 1 1.0 2 0.8 3 0.8
147. LOAD COMB 9 1.0DL+0.80 (FL+ WLZ)
148. 1 1.0 2 0.8 4 0.8
149. LOAD COMB 10 1.0DL+1.0FL+1.0NLX
150. 1 1.0 2 1.0 5 1.0
151. LOAD COMB 11 1.0DL+1.0FL+1.0NLZ
152. 1 1.0 2 1.0 6 1.0

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153. LOAD COMB 12 1.0DL+1.4FL
154. 1 1.0 2 1.4
155. LOAD COMB 13 1.4DL+1.4FL
156. 1 1.4 2 1.4
157. LOAD COMB 14 1.0DL+1.4FL+1.4WLX
158. 1 1.0 2 1.4 3 1.4
159. LOAD COMB 15 1.0DL+1.4FL+1.4WLZ
160. 1 1.0 2 1.4 4 1.4
161. LOAD COMB 16 1.0DL+1.4FL+1.4NLX
162. 1 1.0 2 1.4 5 1.4
163. LOAD COMB 17 1.0DL+1.4FL+1.4NLZ
164. 1 1.0 2 1.4 6 1.4
165. LOAD COMB 18 1.0DL-1.4FL+1.4WLX
166. 1 1.0 2 -1.4 3 1.4
167. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 42/ 70/ 8

SOLVER USED IS THE IN-CORE ADVANCED SOLVER

TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 228

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 1

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 2

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 3

**START ITERATION NO. 2

**START ITERATION NO. 3

**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 4

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 5

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 6

168. LOAD LIST 7 TO 11
169. PRINT JOINT DISPLACEMENTS ALL

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
2	7	0.0000	0.0000	0.0000	-0.0022	0.0013	0.0006
	8	0.0000	0.0000	0.0000	-0.0017	0.0011	0.0006
	9	0.0000	0.0000	0.0000	-0.0023	0.0010	0.0005
	10	0.0000	0.0000	0.0000	-0.0022	0.0014	0.0007
	11	0.0000	0.0000	0.0000	-0.0029	0.0013	0.0006
5	7	0.0000	0.0000	0.0000	-0.0025	-0.0015	-0.0002
	8	0.0000	0.0000	0.0000	-0.0020	-0.0014	-0.0001
	9	0.0000	0.0000	0.0000	-0.0026	-0.0012	-0.0001
	10	0.0000	0.0000	0.0000	-0.0025	-0.0017	-0.0002
	11	0.0000	0.0000	0.0000	-0.0032	-0.0015	-0.0002
7	7	0.0000	0.0000	0.0000	-0.0024	0.0013	0.0007
	8	0.0000	0.0000	0.0000	-0.0019	0.0011	0.0007
	9	0.0000	0.0000	0.0000	-0.0027	0.0010	0.0006
	10	0.0000	0.0000	0.0000	-0.0024	0.0014	0.0008
	11	0.0000	0.0000	0.0000	-0.0032	0.0013	0.0007
8	7	0.0000	0.0000	0.0000	0.0000	0.0012	0.0007
	8	0.0000	0.0000	0.0000	0.0000	0.0011	0.0007
	9	0.0000	0.0000	0.0000	-0.0007	0.0010	0.0006
	10	0.0000	0.0000	0.0000	0.0000	0.0014	0.0008
	11	0.0000	0.0000	0.0000	-0.0004	0.0012	0.0007
9	7	0.0000	0.0000	0.0000	0.0002	0.0011	0.0008
	8	0.0000	0.0000	0.0000	0.0002	0.0010	0.0007
	9	0.0000	0.0000	0.0000	-0.0004	0.0009	0.0006
	10	0.0000	0.0000	0.0000	0.0002	0.0013	0.0009
	11	0.0000	0.0000	0.0000	-0.0002	0.0011	0.0008
13	7	0.0000	0.0000	0.0000	-0.0027	-0.0016	-0.0002
	8	0.0000	0.0000	0.0000	-0.0022	-0.0014	-0.0001
	9	0.0000	0.0000	0.0000	-0.0030	-0.0012	-0.0001
	10	0.0000	0.0000	0.0000	-0.0027	-0.0017	-0.0002
	11	0.0000	0.0000	0.0000	-0.0035	-0.0016	-0.0002
18	7	0.0000	0.0000	0.0000	-0.0005	-0.0016	-0.0002
	8	0.0000	0.0000	0.0000	-0.0004	-0.0014	-0.0001
	9	0.0000	0.0000	0.0000	-0.0012	-0.0013	-0.0001
	10	0.0000	0.0000	0.0000	-0.0005	-0.0018	-0.0002
	11	0.0000	0.0000	0.0000	-0.0009	-0.0016	-0.0002
23	7	0.0000	0.0000	0.0000	-0.0002	-0.0014	-0.0001
	8	0.0000	0.0000	0.0000	-0.0002	-0.0013	-0.0001
	9	0.0000	0.0000	0.0000	-0.0008	-0.0011	-0.0001
	10	0.0000	0.0000	0.0000	-0.0002	-0.0016	-0.0002
	11	0.0000	0.0000	0.0000	-0.0006	-0.0014	-0.0001
25	7	-0.7624	-0.0787	0.0043	0.0004	-0.0007	0.0020
	8	-0.6783	-0.0788	0.0046	0.0005	-0.0006	0.0018
	9	-0.6094	-0.0705	-0.2017	0.0004	-0.0006	0.0016
	10	-0.9089	-0.0786	0.0039	0.0004	-0.0007	0.0024
	11	-0.7620	-0.0729	-0.1295	0.0004	-0.0007	0.0020
27	7	-0.7638	-0.0891	-0.0747	0.0005	-0.0007	0.0020
	8	-0.6795	-0.0890	-0.0594	0.0005	-0.0006	0.0018
	9	-0.6106	-0.0772	-0.3335	0.0004	-0.0006	0.0016